

For SOFR–ASW, most of the differences do not arise also from the unsecured–secured basis anymore, but only from the default risk of the bond, which typically increases with its maturity. If this difference is high, i.e. in case of riskier bond issuers, the difference between swap and bond yield curves can be minimized by using risk-free bond yields, as described in Chapter 13. Then, global bond RV can be assessed via the SOFR–ASW of asset&basis swapped risk-free bond yields, i.e. by using the bond yield minus the adjusted CDS as input variable.

If USD SOFR is used as a reference rate and a yardstick for comparisons, the par asset swap spreads of asset&basis swapped bonds (if necessary after subtracting the adjusted CDS) are natural, relevant, and universal rich/cheap indicators. In this case, there exists therefore an alternative to using fitted curves for assessing the RV between different bonds – which offers the additional benefit of assessing the RV of every bond also versus the CDS, as described in Chapter 16. We therefore transfer the method of external explanatory variables developed for fitted curves (Chapter 8) to this alternative. On the other hand, the same transition from LIBOR to SOFR, which has alleviated the deficiencies of swap spreads as rich/cheap measures has also caused a structural break in the time series of swap spreads. Thus, while fitted curve models may be replaced by SOFR–ASW as RV indicators for Treasuries and all bonds globally swapped into USD, they are of particular importance to create long-term time series for yields (e.g. constant maturity par rates of US Treasuries) as input data for analysis.

If a reference rate other than SOFR is used as a yardstick, the deficiencies of swap spreads as RV indicators are so large that fitted curves remain the only suitable choice for rich/cheap comparisons of individual bonds.¹ This also applies when all bonds are to be compared in a different currency than USD, for example, assessed from the perspective of a EUR-based investor. In this case, every bond should be asset&basis swapped into a note with fixed EUR payments and then be used as input into a fitted curve model (rather than being assessed by its spread to EURIBOR or ESTR).

In summary, there are two suitable alternatives to assess the RV between global bonds:

- For any currency, all bonds can be expressed via a combination of asset&basis swaps as a fixed rate in that currency. Then, the fitted curve model from Chapter 8 can be applied to evaluate the relative pricing between all bonds involved.

¹On the other hand, in countries without transition, there could exist long-term time series of swaps (e.g. with EURIBOR as the reference rate) without structural breaks, hence an alternative to using constant maturity government yields from a fitted curve model as input data.

- For USD, the SOFR–ASW of all bonds can be calculated and used as a global RV indicator (potentially after some adjustments).

This chapter first demonstrates the reasons behind this statement before providing detailed examples for both alternatives.

SOFR SWAP SPREADS AS A GLOBAL RELATIVE VALUE INDICATOR FOR BONDS

A later section of this chapter will demonstrate that the suitability of par asset swap spreads as a RV indicator depends on the proximity of the bond and swap yield curves. Due to the convergence of both yield curves following the transition to SOFR, par SOFR–ASW can therefore be considered to be a good candidate for assessing the richness and cheapness between different bonds – which can be extended to cover all markets via the CCBS.

Taking a closer look at this idea, recall the formula from Chapter 12:

$$\text{SOFR-ASW} = \text{repo-SOFR basis swap} + \text{adjusted CDS}.$$

Thus, the discrepancy between SOFR-swap and bond yield curves would be minimal, if the repo–SOFR basis swap (as well as the haircut financing costs not included in the formula) and the adjusted CDS were zero. In that case, the SOFR–ASW would reflect only the richness and cheapness of a given bond such as being caused by ephemeral flows, not by structural factors such as the credit quality – and hence be an ideal RV indicator indeed.

In order to approach this ideal, we therefore need to eliminate the remaining structural factors from the SOFR–ASW, just as the transition from LIBOR to SOFR has already eliminated the unsecured–secured basis. Using the formula above, we can split this task into two components:

- Bond-specific factors such as the specialness ($\approx$ repo–SOFR basis swap) and other elements not included in the formula above (haircut financing costs, benchmark status, etc.) can be captured via external explanatory variables, as explained in Chapter 8. This corresponds to transferring the technique of adjusting the rich/cheap indicator for structural factors (such as being deliverable) from the fitted curves to SOFR–ASW. And since specialness is one possible external explanatory variable, this approach is suitable to eliminate the repo–SOFR basis swap from the SOFR–ASW. In practical terms, the same multiple regression against a number of external explanatory variables from Chapter 8 can be applied to SOFR–ASW; while the key trick in case of fitted curves consisted in executing this regression simultaneously with the curve fitting, this is not required for SOFR–ASW anymore since the curve is already given.

- The issuer-specific factor of credit risk remains a source of discrepancy between swap and bond yield curves even after the transition to SOFR. As in the formula above, it can be captured via the adjusted CDS and eliminated from SOFR–ASW by using bond yields minus the adjusted CDS as input variables. This corresponds to comparing bonds on the basis of their risk-free yield, as described in Chapter 13. Since the purpose is analytic, calculating risk-free yields by subtracting the adjusted CDS (not the full CDS quotes observed in the market) seems to be preferable.

For most bonds of issuers with high credit quality, one often finds that both structural factors are rather small: only a few bonds have significant specialness and the (adjusted) CDS for the US or Germany are at times (though not always) below 10 bp. In these situations, one may decide that the benefits of eliminating these factors are in practice not worth the effort. However, as the credit risk increases, so does the difference between swap and bond yield curves and hence also the importance of its elimination. This is particularly the case when including bonds of issuers with a higher default risk in the global RV comparison. And since the CDS tends to increase with the maturity, it is also the precondition for meaningful comparisons of the RV between bonds of different maturities: while the cheapest 2Y Treasury may be identified by simply looking at the SOFR–ASW, when comparing individual 2Y with 10Y Treasuries, the higher default risk of the US over 10Y should be included. It therefore seems advisable to generally base the rich/cheap analysis on SOFR–ASW with eliminated structural factors.

PROBLEMS WITH THE USE OF SWAP SPREADS AS RELATIVE VALUE INDICATOR FOR BONDS

The main objective of this section is the demonstration of the statement above that the difference between swap and bond yield curves causes deficiencies of swap spreads as rich/cheap measures. Simultaneously, different categories of swap spreads are briefly introduced and some outstanding technical points (such as bonds trading away from par, which have been abstracted in the conceptual discussion of the previous chapters), are addressed.

Interpolated Swap Spreads

Some investors are concerned that the comparison of the bonds via par asset swaps is complicated by the fact that the two par asset swaps involve up-front payments of different sizes by the swap dealer. For example, if a 5% issue is trading at par, no up-front payment will be required, while a 10% issue is likely to involve a substantial up-front payment from the dealer.

In order to mitigate any problems that might result from these differing up-front payments, some investors prefer to compare the yield of a bond to the

par swap rate with the same maturity as the bond. This difference generally is referred to as the interpolated swap spread, since the maturity of the bond is unlikely to fall on a yearly swap point quoted in the market, and therefore the corresponding swap rate will need to be interpolated. In the example above, let's assume the yield of the 10% bond is 4.98%, the yield of the 5% bond is 5%, and the swap rate with the same maturity as these bonds is 5.35%. In this case, the interpolated swap spreads of the 10% and 5% bonds are -37 bp and -35 bp, and the 10% issue appears to be the richer of the two issues.

Notice in this example that we could have reached the conclusion that the 10% issue is richer than the 5% issue simply by comparing their yields, since the two bonds in this example are assumed to have the same maturity dates. Of course, this would not be the case more generally.

The problem with the interpolated swap spread as a relative value indicator for bonds is the easiest to identify, so we start with this case. In the example of the 5% and 10% bonds above, let's assume the yield curve is upward-sloping. Let's also assume that both bonds are fairly valued to the government bond curve. For example, this would be the case if these two bonds, which we assume to have identical maturity dates, were strippable and there was no reconstitution arbitrage, even under the assumption of zero bid/ask spreads for the bonds and their strips.

Under these assumptions, it can be shown mathematically that the 10% bond has to have a lower yield than the 5% bond. Intuitively, this result is due to the fact that the 10% issue has more of its cash flows discounted at relatively lower rates, since it has the greater coupon. As a result, the 10% bond has to have the wider interpolated swap spread. In this case, it will always appear to be the richer bond, according to the interpolated swap spread, even though both bonds are fairly valued in this example by construction. As this simple example demonstrates, the interpolated swap spread is a fundamentally flawed measure of relative value between two bonds.

Par Asset Swap Spreads

The par asset swap spread seems both to address the problem of matching the principal cash-flows in the CCBS via the up-front payment as described below and to be an ideal measure of relative value between bonds: since the bond with the more negative par asset swap spread will pay the investor less for the same initial investment, it should be viewed as the richer bond. For example, consider again two bonds with identical maturity dates but with different coupons, say, 5% and 10%. If the 10% issue is trading at a par asset swap at LIBOR -40 basis points (bp) and the 5% bond is trading at a par asset swap spread of LIBOR -35 bp, the 10% bond would appear to be the richer bond. For the same initial investment of USD 100, the payoff of the 10% bond is lower than that of the 5% bond. The clean structure of the par asset swap spread makes such comparisons relatively straightforward.

The par asset swap is a cash-flow matched structure. As a result, the par asset swap spread doesn't suffer from the same defect as the interpolated swap spread. However, it does suffer from a more subtle problem relating to the coupon, which can be seen by conducting a thought experiment. Let's start the experiment by assuming the 5% bond has a perfectly fair value, in the sense that there is no reconstitution arbitrage between it and the strips in the market, even assuming zero bid/ask spreads. Now let's increase the coupon of the 5% bond incrementally, until it becomes a 10% bond, and let's assume that the bond continues to have a perfectly fair value to the strips throughout the experiment. We'll watch the behavior of the par asset swap spread as the coupon of the bond is increased.

When the coupon of the bond is increased by 10 bp, from 5% to 5.10%, the price of the bond also increases. Let's assume the bond price increases by 70 cents, from 100.00 to 100.70. In order to see the impact, if any, on the par asset swap spread, let's review the way this spread is calculated. The par asset swap spread is calculated by discounting the cash-flows of the bond to determine a present value for the bond using the swap curve as the source of the discount rates. Let's call this present value the bond's "swap curve present value". The par asset swap spread is the size of an annuity that has the same present value as the difference between the bond's market price and its swap curve present value.

If the term structure of swap rates is uniformly greater than the term structure of government bond yields, a bond's market price will increase by more than a bond's swap curve present value as we increase the coupon of the bond, since the swap curve will involve higher discount rates and lower discount factors than will the government bond curve. In our example, the bond price increased by 70 cents when we added 10 bp to the coupon, but the swap curve present value would increase by a lesser amount, say, 65 cents. As a result, the difference between the market price of the bond and its swap curve present value increased by 5 cents. Therefore, the present value of the par asset swap spread as an annuity must increase by 5 cents. Since the swap curve doesn't change in this example, the only way for this present value to change is for the size of the par asset swap spread to increase. In other words, increasing a bond's coupon will increase its par asset swap spread, even if the bond is assumed to have a perfectly fair value as measured by the government bond curve. The necessary conclusion from this example is that two fairly valued bonds almost always will have different par asset swap spreads if they have different coupons, even if they have the same maturity date. Note that this coupon effect is not the same as the coupon effect we discussed above. The "yield" coupon effect is not an issue when the yield curve is flat, but this "par asset swap spread" coupon effect is an issue even if the yield curve is flat. The only time this par asset swap spread coupon effect isn't an issue is when the swap curve is identical to the government bond curve.

And, in fact, with the transition from LIBOR to SOFR eliminating the unsecured–secured basis as one major source of differences between swap and bond yields and the remaining sources can be relatively easily addressed by the methods described above, SOFR–ASW comes close to the ideal situation of identical swap and government bond curves. Thus, par asset swaps spreads with SOFR as reference rate (and with the modifications described above) represent suitable RV indicators, reflecting only ephemeral imbalances from flows rather than structural factors. By contrast, par asset swap spreads against a different reference rate (e.g. EURIBOR) suffer from the difference between the swap and bond yield curve including the unsecured basis.

In summary, par asset swap spreads are more useful as rich/cheap indicators when the swap curve and the bond curve are closer to one another. With SOFR as the reference rate, the proximity is usually sufficient or can be made sufficient by the techniques described above; for other reference rates, the distance remains too large to use par asset swap spreads as a suitable RV measure.

Full Asset Swap Spreads

While some investors use the interpolated swap spread to mitigate possible difficulties with the par asset swap spread, other investors will use the full asset swap spread. Like the interpolated swap spread, the full asset swap spread involves no up-front payments. Like the par asset swap, the full asset swap is a cash-flow matched structure (i.e. the fixed cash-flow dates and amounts in the swap are identical to the coupon dates and the coupon size of the bond).

To see that the full asset swap spread suffers from a similar defect (in case of a difference between swap and bond yield curves) to the par asset swap spread, recall that the full asset swap spread is equal to the par asset swap spread multiplied by par and divided by the dirty price of the bond.² As basis points are added to the bond's coupon, the par asset swap spread increases by a constant amount per basis point. The full asset swap can only remain unchanged in the thought experiment above if the change from par asset swap spread to full asset swap spread involved the par asset swap spread decreasing by the same constant amount per basis point. Since this is not the case, the full asset swap spread also changes as basis points are added to the bond's coupon in case of the swap curve not matching the bond yield curve.

Zero-Coupon Asset Swap Spreads

So far, we've focused on coupon effects as the sources of concern in using swap spreads for relative value comparisons between bonds. Since zero-coupon bonds are free from such problems, it's worth considering whether the relative

²The dirty price is the clean price plus accrued interest.

values between zero-coupon bonds can be determined by comparisons to the swap curve. For example, let's assume we have two zero-coupon government bonds, a 10-year (10Y) bond with a -50 bp swap spread and a 9Y bond with a -45 bp swap spread, assuming each swap spread is computed as the difference between the bond's yield and the zero-coupon rate with the same maturity as the bond calculated using the swap curve. Would we be justified in claiming the 10Y zero-coupon bond was rich to the 9Y bond?

In case of using non-SOFR swap spreads such as LIBOR-ASW, the answer is an unequivocal no, again due to the inclusion of the unsecured-secured basis. As we saw in Chapter 11, the fair LIBOR-ASW is a function of expected LIBOR-repo spreads, and there will be a presumed forward LIBOR-repo spread that will cause both the 9Y and the 10Y swap spreads to appear to be fair value. The only time at which we might be tempted to use the relative swap spreads as a guide to valuation is if the forward LIBOR-repo spread was negative (i.e. the forward repo rate was greater than the forward LIBOR). Unless there are special circumstances involved, such as different tax treatments or leverage constraints (see Chapter 18), the unsecured LIBOR should never be less than the secured repo rate and the LIBOR-repo spread should never be negative, even on a forward basis.

Another problem with this approach is that the spread between LIBOR and the repo rate should be a function of the level of interest rates. Even if the default probability implied by LIBOR remained the same, the LIBOR-repo spread would be expected to increase as the level of the repo rate increases, as discussed in Chapter 11.

If the 1Y rate nine years forward is higher than the present-value weighted averages of the first nine 1Y forward rates along the curve, then the fair 1Y LIBOR-repo spread nine years forward will also be larger than the first nine 1Y LIBOR-repo spreads along the curve. In this case, the fair zero-coupon swap spread for a 10Y bond will be greater than the fair zero-coupon swap spread for a 9Y bond. From this example, we see that a simple, unadjusted comparison of zero-coupon swap spreads versus LIBOR is inappropriate to assess the relative values between zero-coupon bonds.

CONCLUSION

The larger the discrepancy between swap and bond yield curves, the less suitable swap spreads are as RV indicators.

The transition from LIBOR to SOFR has eliminated the unsecured-secured basis as one source of discrepancy between swap and bond yield curves. The remaining sources can be relatively easily eliminated as well (and are usually of a smaller size). Hence, par asset swap spreads versus SOFR are a suitable RV indicator if used following the approach outlined above.

On the other hand, non-SOFR-ASW incorporates the unsecured-secured basis, which cannot easily be eliminated due to it depending on the variable cost of capital, which is unobservable *ex ante*. Hence, it seems impossible to obtain a suitable RV indicator from LIBOR-ASW or OIS-ASW. This may appear to be a rather strong conclusion given that many analysts have applied and still apply LIBOR-ASW as RV indicators, but is nevertheless a necessary conclusion.

This conclusion also means, that in all currencies apart from USD there exist no swap spreads (see Table 11.1) as a suitable alternative to rich/cheap assessments via fitted curves, as outlined in Chapter 8. If the RV of global bonds shall be compared not in USD but in another currency, it therefore also needs to be done via fitted curves (rather than via swap spreads). For example, if a EUR-based investor wants to select the cheapest bond in the 5Y sector between Bunds and Treasuries, he needs to compare both via a fitted curve model. The precondition for being able to do so is to asset&basis swap the Treasury into a bond with *fixed* EUR cash flows (coupons), just like the Bunds. When comparing all bonds in USD, their SOFR-ASW, i.e. their spread versus a floating rate, can be used; when comparing all bonds in another currency, their fixed rate needs to be calculated as input variable for a fitted curve model.

The next two sections describe the technicalities of both alternatives for assessing relative value between government bonds in different currencies.

Rich/Cheap Analysis through Fitted Curves for Bonds Denominated in Different Currencies

The general approach is straightforward, consisting of three main steps:

1. Express all bonds as fixed rate cash-flows in the same currency, using asset swaps, ICBS, and CCBS as needed.
2. Fit a benchmark yield curve.
3. Calculate the extent to which bonds are rich or cheap relative to this benchmark fitted yield curve.

Turning to the details of the first step, the key is to ensure matching cash-flows between all the transactions, specifically:

- The notional of the CCBS is set to equal the principal at maturity. As the dirty price of the bond is usually not equal to 100, the CCBS should be connected with a *par* asset swap, whose up-front payment equals the difference between the dirty price and par. One can think of buying the bond and par asset swapping it in local currency first and then subject the whole package to a CCBS transaction. In this way, one conceptually deals with a principal cash-flow of 100 both at the start and at the end of the CCBS.

- The different swaps involved should have the same settlement conventions.³ For example, if the asset swap used T+0 and the CCBS T+2, the 2-day mismatch would result in unwanted exposure. At the time of writing (October 2023), it seems feasible (even standard in most currencies) to use the convention T+2 for all asset&basis swaps involved.

Let's consider the example of swapping the US Treasury 3.375% maturing on May 15, 2033 into a fixed note with GBP cash-flows, using market prices from October 17, 2023 and taking the perspective of a GBP-based investor, who wants to generate GBP cash-flows with credit exposure to the US government:

Step 1.1: Buy the US Treasury and par asset swap it (all in USD). With a clean price of USD 88.94 and accrued interest of 1.43, the dirty price is USD 90.37. Enter into a pay fixed/receive floating par asset swap, i.e. agree to pay the semi-annual coupons $3.375/2$ every May 17 and Nov 17 up to and including May 17, 2033 (assuming T+2) and the upfront payment of $USD\ 100 - 90.37 = 9.63$, versus receiving SOFR plus 30 bp (the par asset swap spread) every May 17 up to and including May 17, 2033 (again, assuming T+2). Note that the convention is to add the swap spread to the *compounded* SOFR, not to the daily SOFR rates before compounding. After canceling out the matching cash-flows, you have to pay USD 100 at settlement versus receiving USD 100 at maturity plus 2 days and will receive USD SOFR plus 30 bp every May 17. Hence, the combination of the bond with a par asset swap results in cash-flows, which can be easily combined with a CCBS in the next step.

Step 1.2: Convert the USD into GBP cash flows via a CCBS. Using the current exchange rate of 1.22, you agree to pay GBP 81.97 versus receiving USD 100 at settlement and to pay USD 100 versus receiving GBP 81.97 on May 17, 2033 (using T+2 also in the CCBS). On every May 17, you pay USD SOFR versus receiving GBP SONIA -19 bp (-19 bp being the current quote for the 10Y SOFR-SONIA CCBS). As a result of these transactions, apart from the principal (GBP 81.97) at settlement and maturity plus two days, you receive every year GBP SONIA -19 bp and the USD par asset swap spread 30 bp, which is left after the USD SOFR cancels out. Converting these 30 bp from USD into GBP is done by multiplying with the conversion factor (the quotient of the basis point values in USD and GBP, see Chapter 16), which is currently almost 1 due to the proximity of the SOFR and SONIA yield curves. Hence, together with the CCBS spread, the yearly payment to you is SONIA + 11 bp. At this stage, you only deal with GBP cash-flows anymore; in fact, you have bought now a notional of GBP 81.97 of a synthetic 10Y floating rate note paying SONIA + 11 bp every May 17 issued by the US government.

³See Chapter 3 of Huggins and Schaller (2022), for more details.

Step 1.3: Reverse asset swap floating into fixed GBP. Staying in the GBP universe, agree to pay SONIA + 11 bp every May 17 versus receiving GBP 2.23 every May 17 and Nov 17 until May 17, 2023. Hence, you are now invested into a notional of GBP 81.97 of a synthetic 10Y US Treasury bond paying you a 4.46% semi-annual coupon in GBP.

This exercise in cash-flow matching allows you to transform every bond into any currency desired. In the example above, a US Treasury with a yield of 4.83% in USD has transmogrified into a synthetic bond with a yield of 4.46% in GBP.

Once the desired subset of the global bond market has been expressed in the desired currency, in a second step, the fitted curve model from Chapter 8 can be applied. A key decision is which bonds to include in the optimization, i.e. the extent of the set *Bonds* in the formula from Chapter 8. For example, if Bunds are to be compared with US Treasuries on a EUR basis, one could either fit the curve through Bunds only or through Bunds *and* basis swapped Treasuries simultaneously. In the first case, one uses the *local* yardstick from and for Bunds also as the *global* yardstick; in the latter case, one obtains a truly *global* yardstick by also including the information from other bond markets in the optimization.

Figure 17.1 shows the richness and cheapness of bonds swapped into EUR (with annual coupon payments) as of October 17, 2023, including only Bunds in the optimization and using no external explanatory variables. This corresponds to looking at global bond markets from the perspective of Bunds and hence of an investor in Bunds who wants to compare them with other alternatives: all bonds are asset&basis swapped into Bund-like bonds, paying annual coupons in EUR, and are priced versus the discount factor curve fitted through Bunds.

Highlighting a few observations:

- For short maturities, USTs, Gilts, and Bunds are mixed, i.e. differentiated more by other factors (such as repo specialness) rather than issuer. As the time to maturity increases, issuer-specific factors become increasingly influential and correspondingly the differentiation by issuer becomes more visible. For 1Y bonds, the default risk is almost negligible, supporting the pricing of USTs, Gilts, and Bunds versus each other. For 5Y bonds, by contrast, both the default risk and haircut schedule result in a differentiation by issuers.
- In all maturity segments, some US Treasuries offer a significant pick-up over Bunds. Hence, if an investor in Bunds is indifferent about the credit difference between the US and Germany (and the haircut differences),